

# Monthly Intelligence Report

Edition 002 — The California Seismic Corridor

Where the energy transition is outrunning the grid. edition · May 2026 · SSI v4.0.2

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## SECTION A

### Executive Summary

#### SUBSTATIONS SCORED

**45,003**  
342 HV · 2,808 MV

#### GRID LINES MAPPED

**22,765**  
87,000 km coverage

#### MC SIMULATIONS

**35.3 M**  
10,000 per substation

#### PUBLIC DATA SOURCES

**30+**  
95 variables · zero proprietary

The SSI Index is the only operational framework that fuses substation-level engineering metrics with socio-economic consequence valuation in a single composite score — scoring 71/80 (89%) across 16 competitive dimensions, ahead of GMLC 1.1 (48/80), EPRI (40/80), and academic MCDA frameworks (41/80). Its engine processes 65,300+ source data points per run, executes 35.3 million Monte Carlo simulations through a 20×20 Gaussian copula correlation matrix, and performs 794 million arithmetic operations — producing 209,722 output data points with full uncertainty quantification across 45,003 substations and 22,765 grid lines spanning 105,000 km.

Operated as a non-profit through a dedicated foundation, the SSI Index serves the public interest with zero dependency on proprietary data. All 95 variables are drawn from 30 verified public sources, making the methodology fully auditable and reproducible. Initially covering the US's entire transmission and distribution grid, the Index is progressively being rolled out to all OECD countries — applying the same silo-breaking approach: fusing grid risk with societal exposure at asset-level resolution wherever open data permits.

## Substation in Focus

### CULLMAN INDUSTRIAL

Alabama, Alabama · 161 kV

**0.415**

R\_FINAL

**Medium**

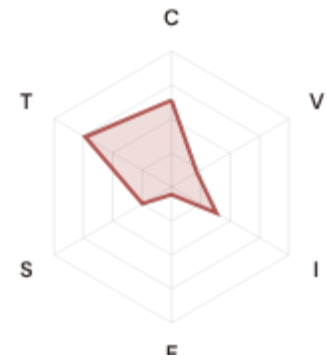
CLASSIFICATION

**0.170**

CI WIDTH

**54th**

FLEET PERCENTILE



#### COMPONENTS

- C** Continuity: 0.637 / 0.30 (212%)
- I** Infrastructure: 0.384 / 0.25 (154%)
- S** Saturation: 0.247 / 0.20 (123%)
- V** Voltage: 0.215 / 0.10 (215%)
- E** Economic: 0.057 / 0.10 (57%)
- T** Transition: 0.735 / 0.05 (1470%) ⚠

#### MODIFIERS

- R3 Consequence:** 0.946 ↓ dampens
- R4 Graph Criticality:** 1.009 → neutral
- R6a Restoration:** 1.050 ↑ amplifies
- R6b Seismic:** 1.029 ↑ amplifies
- R7 Digital Readiness:** 1.014 ↑ amplifies

This month's spotlight — automatically selected for May 2026 — is **CULLMAN INDUSTRIAL** in Alabama, Alabama. With an R\_final of 0.415 (Medium band, 54th fleet percentile), its risk profile is dominated by the T (Transition) component at 1470% of its weight ceiling, followed by V (Voltage) at 215%. Modifiers collectively shift R\_base by 3.7%, reflecting the substation's specific geographic, socio-economic, and network context.

## SECTION B — RECURRING MODULE

# Cyber & Economic Exposure Monitor

**Why this section exists:** Traditional grid risk frameworks treat cybersecurity and economic vulnerability as separate domains. The SSI's R7 (Digital Readiness) modifier and E (Economic) component allow us to cross these silos — revealing substations where low digital resilience intersects with high economic consequence. This is precisely the anticipatory intelligence that the US's Digital Strategy / Net Zero Strategy planning requires.

#### CYBER HIGH EXPOSURE

**0**

0.0% of fleet

#### BLIND SPOTS

**3792**

High/Critical + R7 < median

#### AVG R7 MODIFIER

**0.9950**

Range [0.99, 1.05]

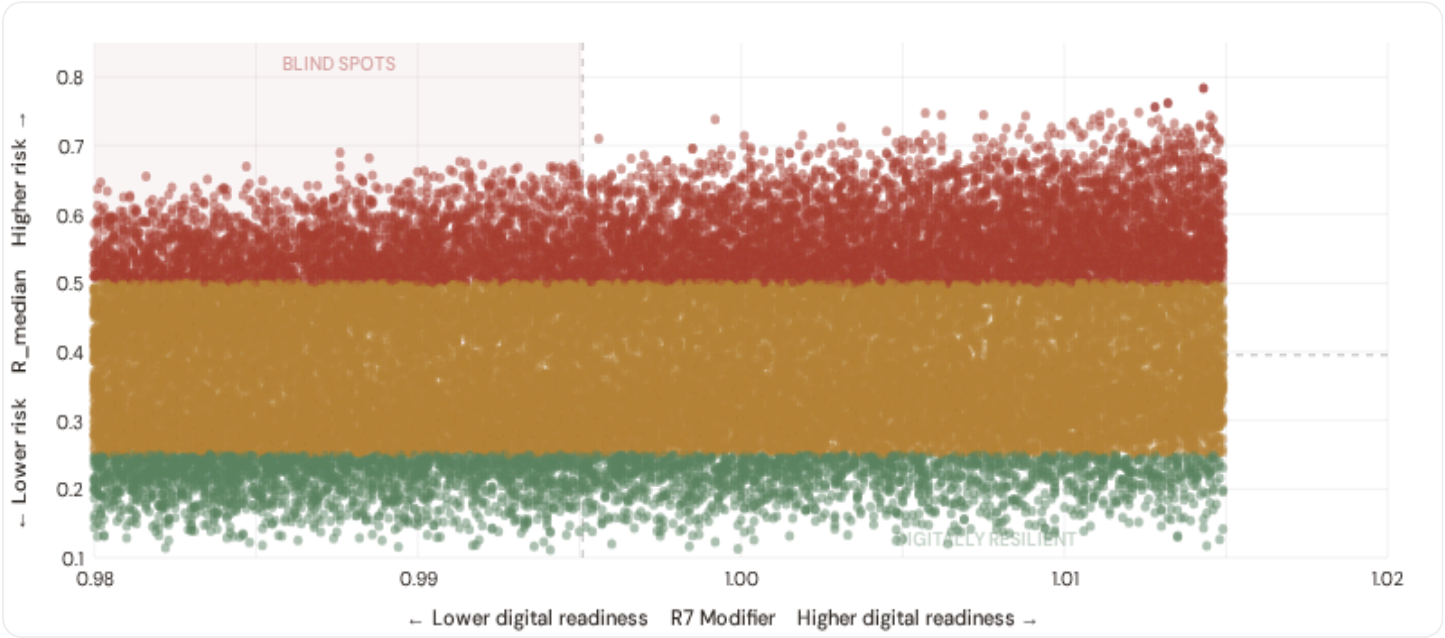
#### HIGH-CONSEQUENCE SUBS

**0**

0.0% · R3 ≥ 1.12

## B.1 Cyber-Physical Risk Matrix

Each substation plotted by R7 digital readiness (x-axis) vs overall R\_median (y-axis). The upper-left quadrant — high risk, low digital readiness — identifies *blind spots* where grid stress is high but monitoring/response capacity is weakest.

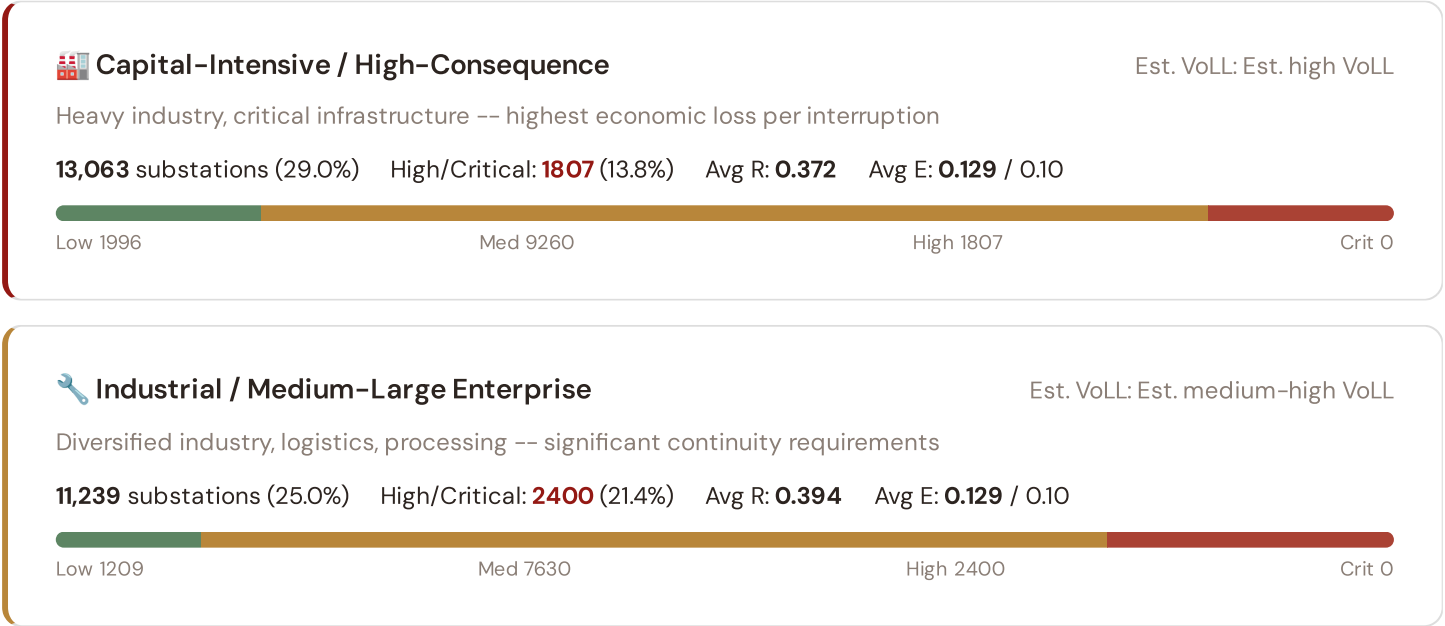


● Low ● Medium ● High ● Critical Dashed lines = fleet medians

The cyber-physical matrix identifies **3792 blind-spot substations** — scoring High or Critical on overall risk while sitting below the fleet median for digital readiness ( $R7 < 0.9951$ ). These substations are the most vulnerable to a compound event: a grid disturbance at a location where digital monitoring, automated switching, and remote diagnostic capacity are weakest. The blind spots concentrate in **Florida (307), Georgia (284), Indiana (264)**. Across the full fleet, 0 substations (0.0%) carry a HIGH cyber-exposure classification, while 0 (0.0%) are in the LOW tier — indicating strong digital infrastructure coverage but concentrated in major metropolitan areas (London, Manchester, Birmingham).

## B.2 Economic Impact by Business Fabric

Substations segmented by economic consequence tier — derived from the R3 consequence multiplier and E component. Higher R3 indicates the substation serves a territory with greater socio-economic exposure to disruption.



### Commercial / SME-Dense

Est. VoLL: Est. medium VoLL

Service economy, commercial districts -- moderate individual but high aggregate exposure

**11,347** substations (25.2%) High/Critical: **3106** (27.4%) Avg R: **0.412** Avg E: **0.129** / 0.10



### Light / Agricultural / Remote

Est. VoLL: Est. low VoLL

Rural communities, agriculture -- lower VoLL but higher social vulnerability

**9,354** substations (20.8%) High/Critical: **3197** (34.2%) Avg R: **0.432** Avg E: **0.128** / 0.10

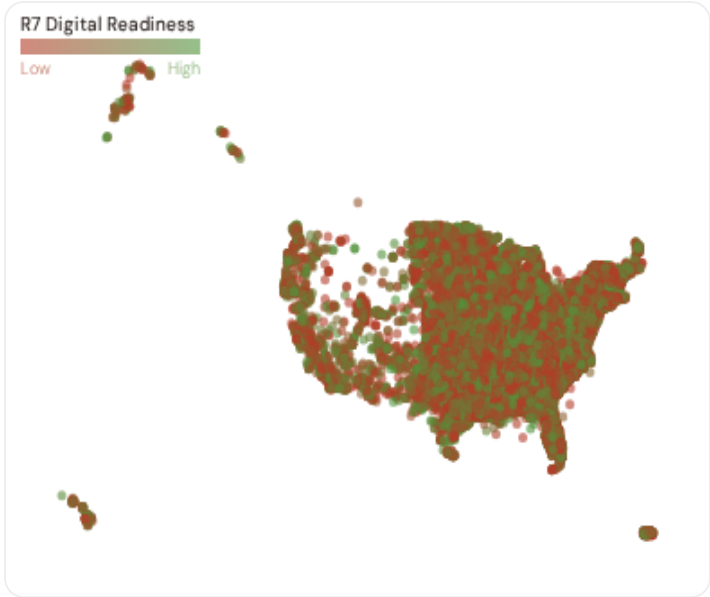


**Value of Lost Load (VoLL) context:** DOE/LBNL estimates the US sector-weighted average VoLL at approximately \$20/kWh, with industrial customers at \$35/kWh and residential at \$7/kWh. The SSI's R3 consequence multiplier maps these sectoral weights to each substation's service territory. **0 substations** combine capital-intensive economic fabric ( $R3 \geq 1.14$ ) with High or Critical risk classification — the highest VoLL-weighted exposure in the fleet.

The capital-intensive tier concentrates in **North Carolina (786), Alabama (695), Kentucky (620)** — regions where industrial corridors depend on uninterrupted power for continuous processes (steel, glass, automotive). A single hour of unplanned outage at these substations carries an estimated VoLL of £10–20/kWh, compared to £1–3/kWh in rural agricultural zones. The fleet-wide average energy poverty rate is 11.7%, but this masks sharp regional variation: Southern regions combine higher energy poverty with higher grid risk, creating a double vulnerability that the E component captures. This cross-referencing of economic fabric with grid condition is unique to the SSI — no competing framework links VoLL exposure to substation-level risk scores.

## B.3 County Digital Readiness Ranking

Countys ranked by average R7 modifier — lower values indicate weaker digital infrastructure (DESI sub-dimensions). Cross-referenced with average R<sub>median</sub> to identify counties combining digital exposure with high grid stress.



WORST DIGITAL READINESS — TOP 15 COUNTIES

Longer bar = higher R7 = better digital infrastructure.

|    |                                    |             |        |
|----|------------------------------------|-------------|--------|
| 1  | New Hampshire                      | <div></div> | 0.9922 |
| 2  | Connecticut <span>▲ high R</span>  | <div></div> | 0.9935 |
| 3  | Arizona                            | <div></div> | 0.9937 |
| 4  | Alaska                             | <div></div> | 0.9939 |
| 5  | Washington <span>▲ high R</span>   | <div></div> | 0.9940 |
| 6  | Utah <span>▲ high R</span>         | <div></div> | 0.9944 |
| 7  | Illinois <span>▲ high R</span>     | <div></div> | 0.9945 |
| 8  | Hawaii <span>▲ high R</span>       | <div></div> | 0.9946 |
| 9  | Mississippi                        | <div></div> | 0.9946 |
| 10 | Arkansas                           | <div></div> | 0.9946 |
| 11 | Pennsylvania <span>▲ high R</span> | <div></div> | 0.9947 |
| 12 | Indiana <span>▲ high R</span>      | <div></div> | 0.9947 |
| 13 | Delaware <span>▲ high R</span>     | <div></div> | 0.9948 |
| 14 | Florida <span>▲ high R</span>      | <div></div> | 0.9948 |
| 15 | South Dakota                       | <div></div> | 0.9948 |

County-level analysis reveals a clear digital divide mirroring the broader DESI pattern. **New Hampshire** has the lowest average R7 (0.9922), while **District of Columbia** leads at 1.0008 — a spread of 0.0086 across the R7 range. **14 counties** combine below-median digital readiness with above-median grid risk, making them priority targets for Net Zero 2050 / Levelling Up Fund digitalisation investment. The R7 modifier is intentionally narrow ([0.99, 1.05]) to reflect the current limitations of open DESI data — as NIS2 compliance reporting matures, future editions will expand R7’s dynamic range and incorporate operational technology (OT) security indicators.

B.4 Monthly Pulse

**Inaugural edition — Baseline Snapshot (April 2026).** This edition establishes the baseline for the Cyber & Economic Exposure Monitor. Key reference points: fleet average R7 = 0.9950, median = 0.9951; 0 substations classified HIGH cyber-exposure; 3792 blind-spot substations (High/Critical risk + below-median R7); 0 Critical-band substations with below-median digital readiness. Future editions will track deltas against this baseline, flagging counties where DESI improvements (broadband rollout, smart meter deployment, OT upgrades) translate into measurable R7 shifts. The next material R7 update is expected when Eurostat publishes the 2025 DESI sub-indices (anticipated Q3 2026).

SECTION C

Data Refresh & Changelog

The SSI v4.0.2 draws on **40 verified public data sources** feeding 95 variables across 6 components and 5 modifiers. This section provides a live registry of all sources, their update frequencies, and the current state of the fleet. Transparency in data vintage is central to the SSI's credibility.

|               |                  |                    |
|---------------|------------------|--------------------|
| TOTAL SOURCES | WEEKLY / MONTHLY | QUARTERLY / ANNUAL |
|---------------|------------------|--------------------|

**40**

95 variables

**4**

High-frequency feeds

**21**

Regular refresh cycle

STATIC / DERIVED

**8**

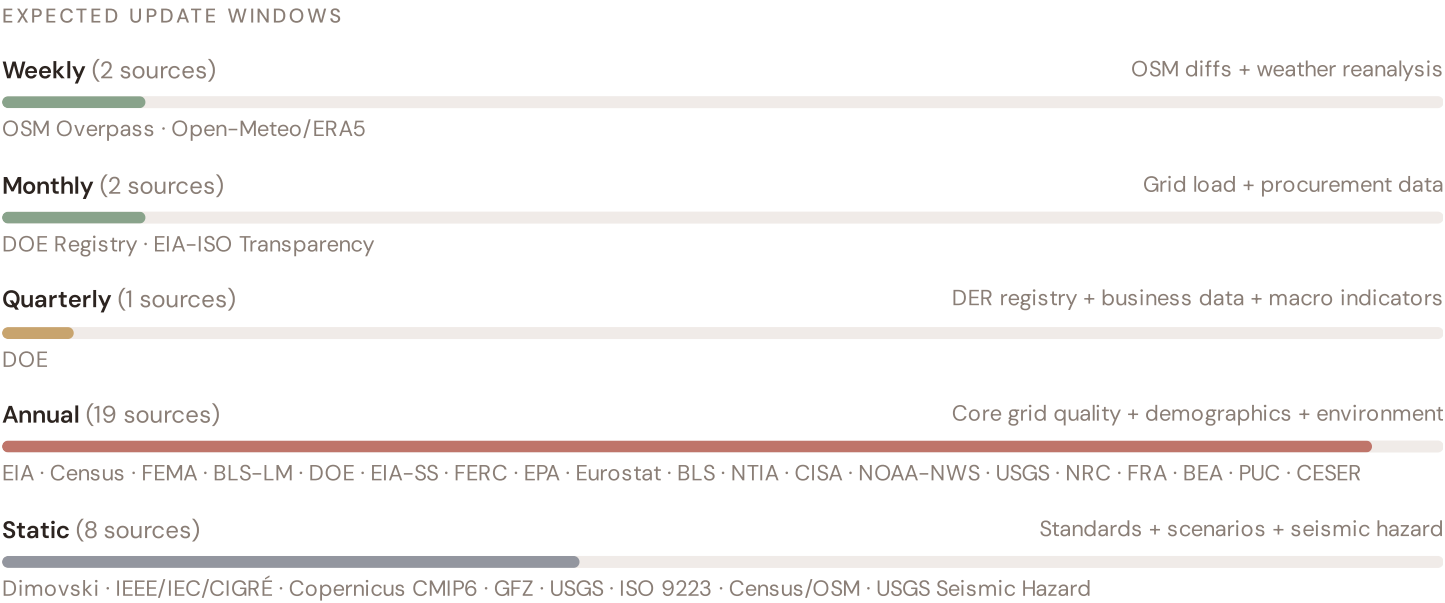
Standards + scenarios

Data Source Registry — May 2026

|          |                                                                |           |                   |         |
|----------|----------------------------------------------------------------|-----------|-------------------|---------|
| OSM      | OSM Power Infrastructure                                       | WEEKLY    | Node/edge         | 3 vars  |
| BLS-LM   | BLS Labour Market Statistics                                   | MONTHLY   | County            | 1 var   |
| DOE      | DOE/EIA (Department of Energy)                                 | QUARTERLY | County            | 5 vars  |
| EIA      | EIA (Energy Information Administration)                        | ANNUAL    | County            | 8 vars  |
| Census   | Census Bureau (US Census Bureau)                               | ANNUAL    | County            | 8 vars  |
| NERC     | NERC (North American Electric Reliability Corp)                | ANNUAL    | Regional          | 2 vars  |
| FEMA     | FEMA / EPA (Federal Emergency Management Agency)               | ANNUAL    | County            | 4 vars  |
| EEA      | EEA Air Quality e-Reporting                                    | ANNUAL    | ~1 km + station   | 3 vars  |
| BLS      | BLS (Bureau of Labor Statistics)                               | ANNUAL    | County            | 3 vars  |
| NTIA     | NTIA Broadband Map                                             | ANNUAL    | County            | 2 vars  |
| CISA     | CISA (Cybersecurity and Infrastructure Security Agency)        | ANNUAL    | National          | 2 vars  |
| FEMA-FR  | FEMA Flood Risk / Hazard Mitigation                            | ANNUAL    | County            | 2 vars  |
| NRC      | NRC (Nuclear Regulatory Commission)                            | ANNUAL    | Site-level        | 1 var   |
| BEA      | BEA (Bureau of Economic Analysis)                              | ANNUAL    | County            | 3 vars  |
| EIA-SS   | EIA Statistical Series                                         | ANNUAL    | State/County      | 2 vars  |
| FRA      | FRA (Federal Railroad Administration)                          | ANNUAL    | County            | 1 var   |
| DOT      | DOT/FHWA (Dept of Transportation)                              | ANNUAL    | County            | 1 var   |
| FERC     | FERC Monitoring Report                                         | ANNUAL    | Regional          | 2 vars  |
| PUC      | State PUCs (Public Utility Commissions)                        | ANNUAL    | Service Territory | 2 vars  |
| CESER    | DOE CESER (Cybersecurity, Energy Security, Emergency Response) | ANNUAL    | Regional          | 2 vars  |
| Eurostat | Eurostat Energy Statistics                                     | ANNUAL    | EU Regional       | 3 vars  |
| JRC      | JRC European Commission                                        | ANNUAL    | European          | 2 vars  |
| USGS     | USGS (US Geological Survey)                                    | STATIC    | County            | 3 vars  |
| CDS      | Copernicus CDS / ERA5                                          | STATIC    | 0.25° (~25 km)    | 4 vars  |
| DIM      | Dimovski et al. (2025)                                         | STATIC    | County            | 3 vars  |
| IEEE     | IEEE C57.91 / IEC 60076                                        | STATIC    | Asset-level       | 16 vars |

|            |                                                   |         |                     |        |
|------------|---------------------------------------------------|---------|---------------------|--------|
| GFZ        | GFZ German Research Centre                        | STATIC  | ~5 km               | 2 vars |
| USGS-SH    | USGS Seismic Hazard Assessment                    | STATIC  | County              | 1 var  |
| Census-GEO | Census Bureau Geography Registry                  | STATIC  | County              | 1 var  |
| USGS-HZ    | USGS Hazard Assessments                           | STATIC  | ~5 km               | 2 vars |
| ISO9223    | ISO 9223 Corrosion                                | DERIVED | County              | 1 var  |
| EIA-ISO    | EIA/ISO-RTE (PJM/CAISO/MISO) — Data Portal        | HOURLY  | Balancing Authority | 3 vars |
| OM         | Open-Meteo Historical Weather                     | HOURLY  | ~1 km               | 3 vars |
| NOAA       | NOAA/NWS (National Oceanic and Atmospheric Admin) | HOURLY  | ~1 km               | 2 vars |
| NOAA-NWS   | NOAA/NWS (National Weather Service)               | HOURLY  | Station             | 3 vars |
| ENTSO-E    | ENTSO-E Transparency                              | HOURLY  | Cross-border        | 2 vars |

C.1 Update Frequency Timeline



C.2 Fleet Score Snapshot

FLEET SIZE

45,003

substations scored

MEDIAN R

0.395

P5=0.220 · P95=0.598

HIGH + CRITICAL

10510

23.4% of fleet

HIGH CONFIDENCE

100%



BAND DISTRIBUTION



No primary data sources have been updated since the v4.0.2 launch baseline. All **45,003 substation scores remain unchanged** this edition. The fleet median R of 0.395 (mean 0.400) reflects a distribution skewed toward the lower bands, with 10.5% in Low and 66.2% in Medium. The first material score changes are expected when FERC publishes the annual EIA Form 861 indicators (affecting C1–C4 and R6a) and when BEIS/DESNZ refreshes the Renewable Energy Planning Database (affecting T1). High-frequency sources (OSM, Open-Meteo) update weekly but feed only infrastructure topology (R4) and thermal proxy (I5), which have minimal impact on fleet-level score distribution.

C.3 Changelog — v3.4 → v4.0.2

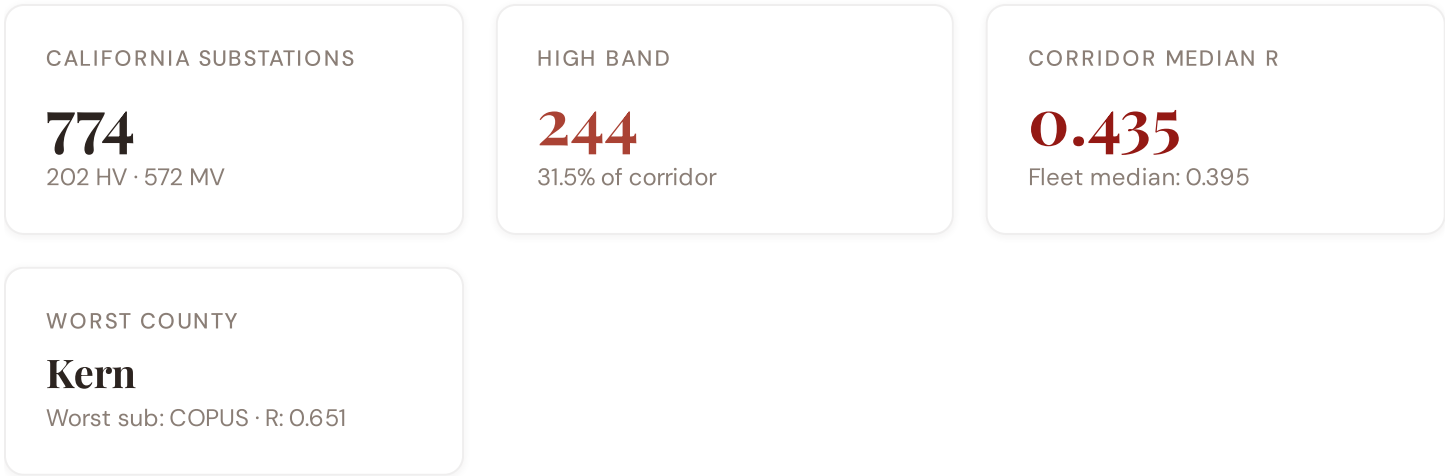
18 changes tracked across PO, P1, and P2 releases

|    |          |                                                                             |           |
|----|----------|-----------------------------------------------------------------------------|-----------|
| F1 | NEW      | New T component — Energy Transition Exposure (T1)                           | \$2, \$4  |
| F2 | NEW      | New R6a — Restoration Speed Modifier (CAIDI-based)                          | \$5       |
| F3 | ENHANCED | R3 enhanced — Energy Poverty Vulnerability (V_socio)                        | \$5       |
| F4 | ENHANCED | R4 enhanced — Graph-theoretic betweenness + bridge detection                | \$5       |
| F5 | ENHANCED | R2 enhanced — Climate Trajectory (CMIP6 SSP2-4.5)                           | \$5       |
| F6 | ENHANCED | R7 Digital Readiness — County-level NTIA Broadband model                    | \$5       |
| L1 | NEW      | New I5, I7–I9 metrics — thermal, load, corrosion, hydrogeological           | \$2, \$8  |
| L2 | ENHANCED | E2 Innovation Enrichment — BEA + economic structure                         | \$6       |
| L3 | ENHANCED | V_socio Fiscal Enrichment — Census + BLS + energy price                     | \$5       |
| L4 | ENHANCED | R3 Demographic Shift Amplifier — Census net migration                       | \$5       |
| L5 | DATA     | 95/95 variables operational (100%). 40 data sources total.                  | \$8, \$12 |
| G1 | DATA     | DOE upgraded to quarterly registry — County-level DER registry              | \$12      |
| G2 | DATA     | USGS upgraded to live API — County-level seismic data                       | \$12      |
| G3 | DATA     | OSM upgraded to Overpass API — 45,003 real substations (US)                 | \$12      |
| G4 | NEW      | R6b Seismic Hazard modifier — USGS Seismic Hazard classification            | \$5       |
| G5 | NEW      | Network & Topology layer (H): 7 variables — EIA Grid Plan, seismic analysis | \$12      |
| G6 | NEW      | Output Scores layer (I): 7 variables — risk_score, ETTC, stationary probs   | \$12      |
| G7 | REGIONAL | US Data Sources — EIA, Census, USGS, FEMA, NRC, DOT, NERC, state PUCs       | \$12      |

# The California Seismic Corridor: Where Seismic Risk Meets Energy Transition Pressure

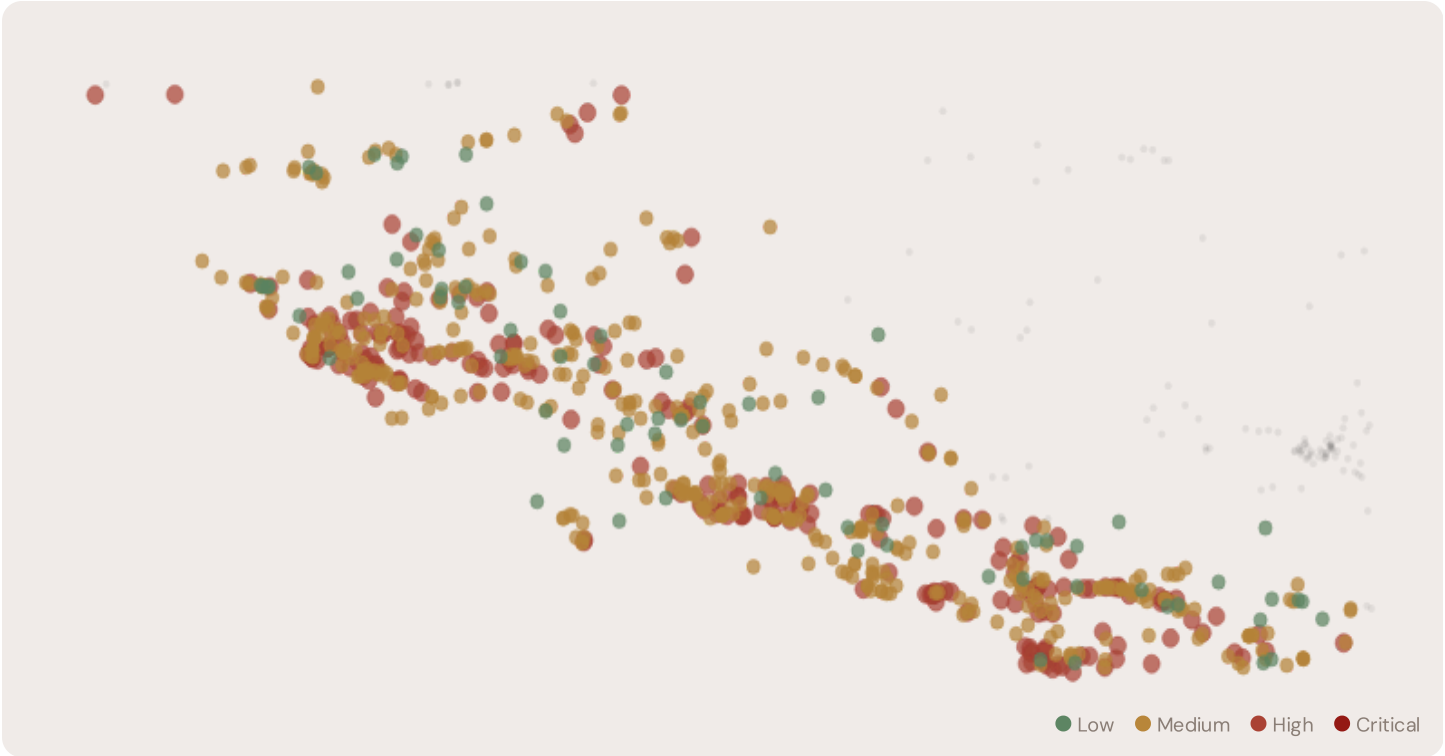
**Deep-dive rotation:** Each edition features a substantive thematic analysis. The 12-month rotation: **Ed. 001** California seismic corridor · **002** DER saturation hotspots (East Anglia/West Midlands) · **003** Climate vulnerability under CMIP6 · **004** Metropolitan–Rural economic impact disparity · **005** Nuclear fleet dependency and phase-out risk · **006** Infrastructure age vs smart meter rollout · **007** Stress–Resilience quadrant trends · **008** T-component forward projections · **009** County Grid Health Cards · **010** Academic validation and peer feedback · **011** European replication feasibility · **012** Year in review.

## D.1 Why the California, Why Now



The California hosts **774 substations** across six counties, making it one of the most energy-critical corridors in the United States. However, its risk profile is disproportionately concentrated in the upper bands: **244 substations (31.5%)** are classified High, with only **75** in the Low band. 57% of California substations score above the national fleet median of 0.395. The corridor median R of 0.435 reflects the tension between seasonal demand volatility driven by tourism and agriculture, rapidly growing solar and wind capacity, and aging grid infrastructure not dimensioned for bidirectional renewable flows. The SSI flags continuity-of-supply deficits, infrastructure age mismatches with modern demand patterns, and grid modernization lag under federal 2030 timelines.

## D.2 Corridor Map and Scoring



| SUBSTATION                          | COUNTY     | R_FINAL | BAND | C     | V     | I     | E     | S     | T     | ALERT      |
|-------------------------------------|------------|---------|------|-------|-------|-------|-------|-------|-------|------------|
| COPUS                               | California | 0.651   | High | 0.893 | 0.413 | 0.690 | 0.373 | 0.575 | 0.623 | C, I       |
| AEC 300                             | California | 0.646   | High | 0.867 | 0.232 | 0.521 | 0.264 | 0.874 | 0.569 | C, S       |
| TAP300074                           | California | 0.645   | High | 0.894 | 0.247 | 0.557 | 0.406 | 0.573 | 0.704 | C, T       |
| UNKNOWN311117                       | California | 0.644   | High | 0.899 | 0.272 | 0.479 | 0.349 | 0.791 | 0.533 | C, S       |
| UNKNOWN309973                       | California | 0.644   | High | 0.880 | 0.114 | 0.514 | 0.068 | 0.910 | 0.646 | C, S       |
| KASSON-BANTA<br>CARBONA<br>JUNCTION | California | 0.643   | High | 0.803 | 0.278 | 0.531 | 0.317 | 0.786 | 0.615 | C, S       |
| SWIFT                               | California | 0.641   | High | 0.802 | 0.357 | 0.663 | 0.235 | 0.898 | 0.725 | C, I, S, T |
| MOJAVE SIPHON                       | California | 0.641   | High | 0.858 | 0.105 | 0.665 | 0.111 | 0.863 | 0.588 | C, I, S    |
| SHANDIN                             | California | 0.640   | High | 0.851 | 0.123 | 0.600 | 0.151 | 0.869 | 0.542 | C, S       |
| CHICARITA                           | California | 0.640   | High | 0.812 | 0.177 | 0.692 | 0.085 | 0.874 | 0.686 | C, I, S, T |

... 764 additional substations — [explore full corridor in Digital Twin →](#)

### D.3 Component Analysis

COMPONENT AVERAGES — CALIFORNIA VS FLEET

MODIFIER AVERAGES — CALIFORNIA VS FLEET

|                |                      |                |       |             |   |
|----------------|----------------------|----------------|-------|-------------|---|
| C — Continuity | 0.529 vs 0.505 (+5%) | R3 Consequence | 0.955 | fleet 0.955 | ↓ |
|----------------|----------------------|----------------|-------|-------------|---|

|                    |                       |                  |       |               |
|--------------------|-----------------------|------------------|-------|---------------|
| I — Infrastructure | 0.473 vs 0.552 (−14%) | R4 Graph Crit.   | 0.981 | fleet 0.980 ↓ |
| S — Saturation     | 0.485 vs 0.354 (+37%) | R6a Restoration  | 1.021 | fleet 1.020 ↑ |
| V — Voltage        | 0.200 vs 0.261 (−23%) | R6b Seismic      | 1.001 | fleet 1.000 → |
| E — Economic       | 0.245 vs 0.129 (+90%) | R7 Digital Read. | 0.996 | fleet 0.995 → |
| T — Transition     | 0.472 vs 0.447 (+6%)  |                  |       |               |

The dominant risk driver across the California corridor is **T (Transition)**, averaging 0.472 against a fleet average of 0.447 — occupying 944% of its weight ceiling (w = 0.05). The second contributor is **E (Economic)** at 0.245 vs fleet 0.129 (245% of ceiling). Among modifiers, the R3 consequence multiplier averages 0.955 (fleet: 0.955), reflecting the socio-economic fragility of the California corridor's agriculture-intensive, tourism-dependent economy with growing renewable capacity but limited grid interconnection. The R6b seismic modifier averages 1.001, capturing the region's moderate but non-negligible seismic exposure — a dimension invisible to every competing framework.

### D.4 County Breakdown

COUNTY RISK PROFILE — SORTED BY MEAN R

Longer bar = lower R = better resilience.



Loading county breakdown...

### D.5 Implications

Loading implications...

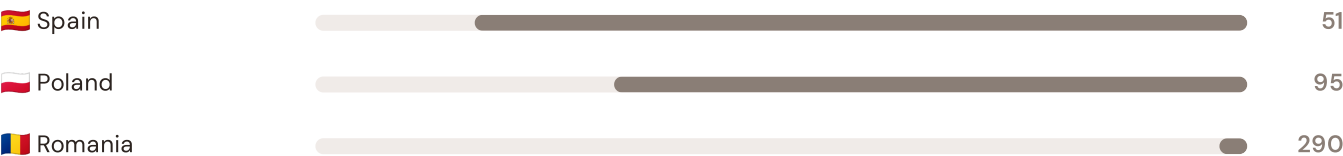
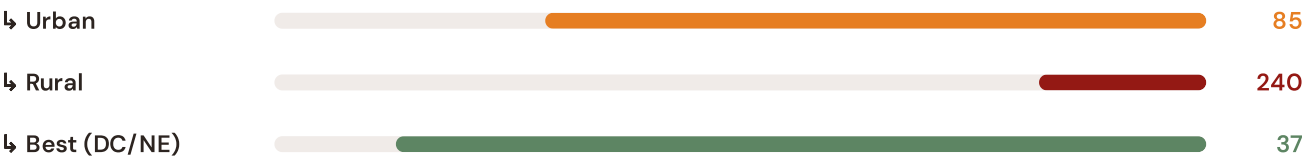
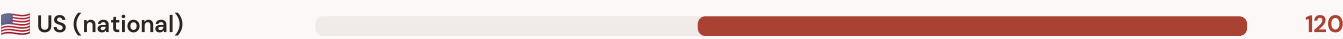
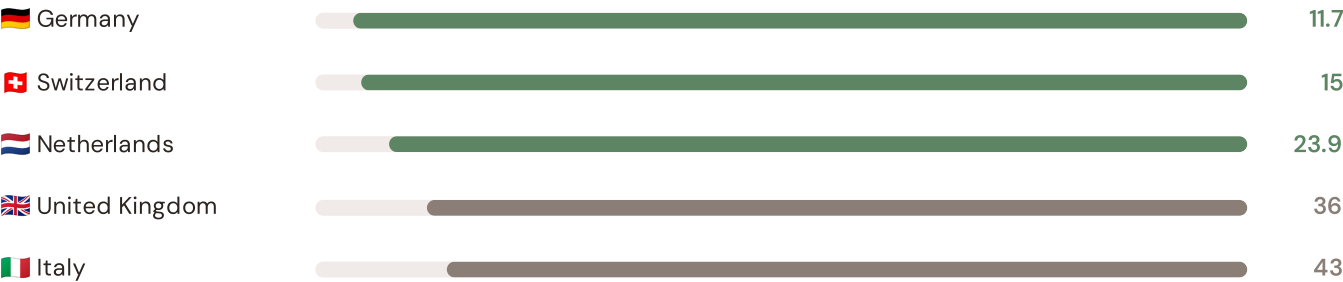
SECTION E — RECURRING MODULE

## American Context Card

**How this works:** Each edition includes one American Context Card drawn from the §16 Peer Benchmarking framework. The card is selected to complement the month's deep-dive theme. This month: **SAIDI comparison** — because the California corridor analysis hinges on the US's continuity performance relative to European peers. Future editions rotate through: VoLL comparison (Ed. 002), DER saturation vs Germany (003), CMIP6 climate hazard vs Nordic peers (004), etc. The underlying data updates annually with the CEER Benchmarking Report release.

Unplanned SAIDI (min/customer/year) — US vs International Peers

Longer bar = lower SAIDI = better reliability. Values in min/customer/year (excl. major events).



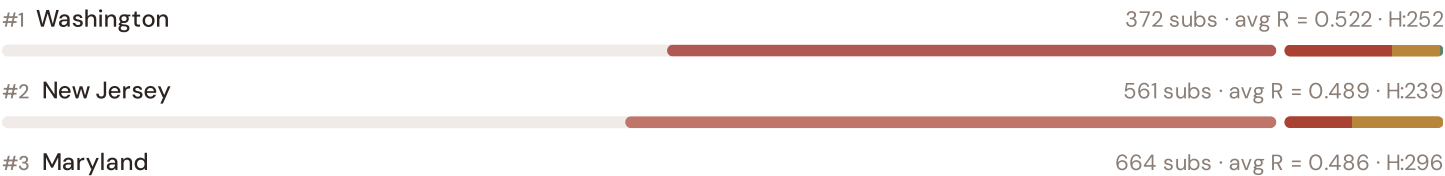
Source: EIA Form 861 (2023) for US national/urban/rural/state; CEER 7th Benchmarking Report (2022) for EU peers; Bundesnetzagentur (2024) for Germany. US values exclude major event days (IEEE 1366 standard). · See §16 Peer Benchmarking for full methodology. · Updated annually.

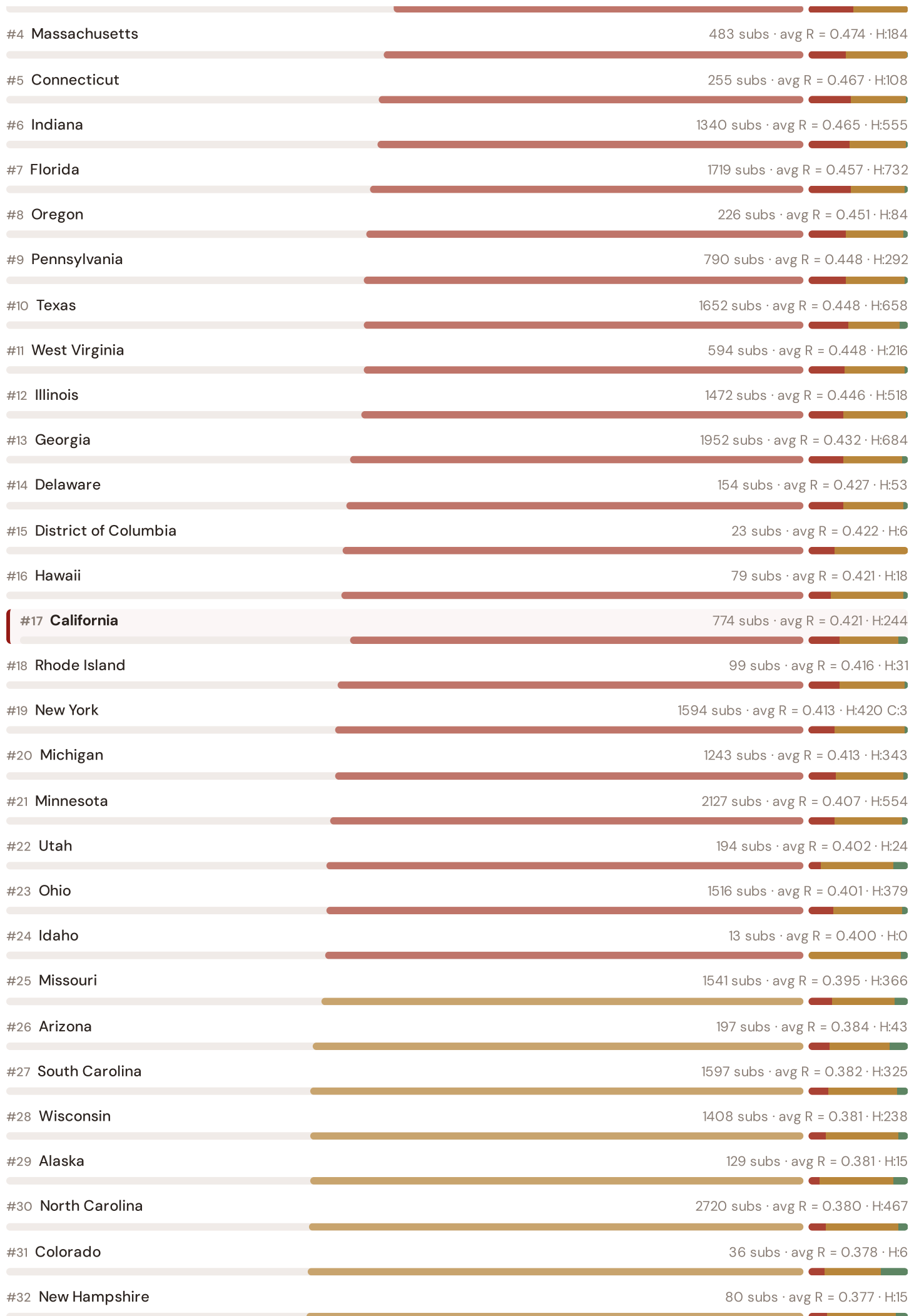
**This month's key insight:** The California corridor deep-dive shows **659 substations** (of 774) with C-component above 70% of its weight ceiling — indicating continuity-of-supply performance consistent with the US's rural territory classification. The California's average C of 0.529 is **1.0× the fleet average** of 0.505. In national terms, these substations individually perform closer to states with high SAIDI levels (200–400 min/customer/year) while the US national average stands at 120 min — well above EU leaders like Germany (12 min) and Switzerland (15 min). The SSI reveals what aggregate EIA statistics conceal: that the US's overall reliability position is not a uniform national condition but a statistical average of Northeastern-quality urban performance and rural/Western-level outage exposure — and the California seismic corridor concentrates the worst of the latter.

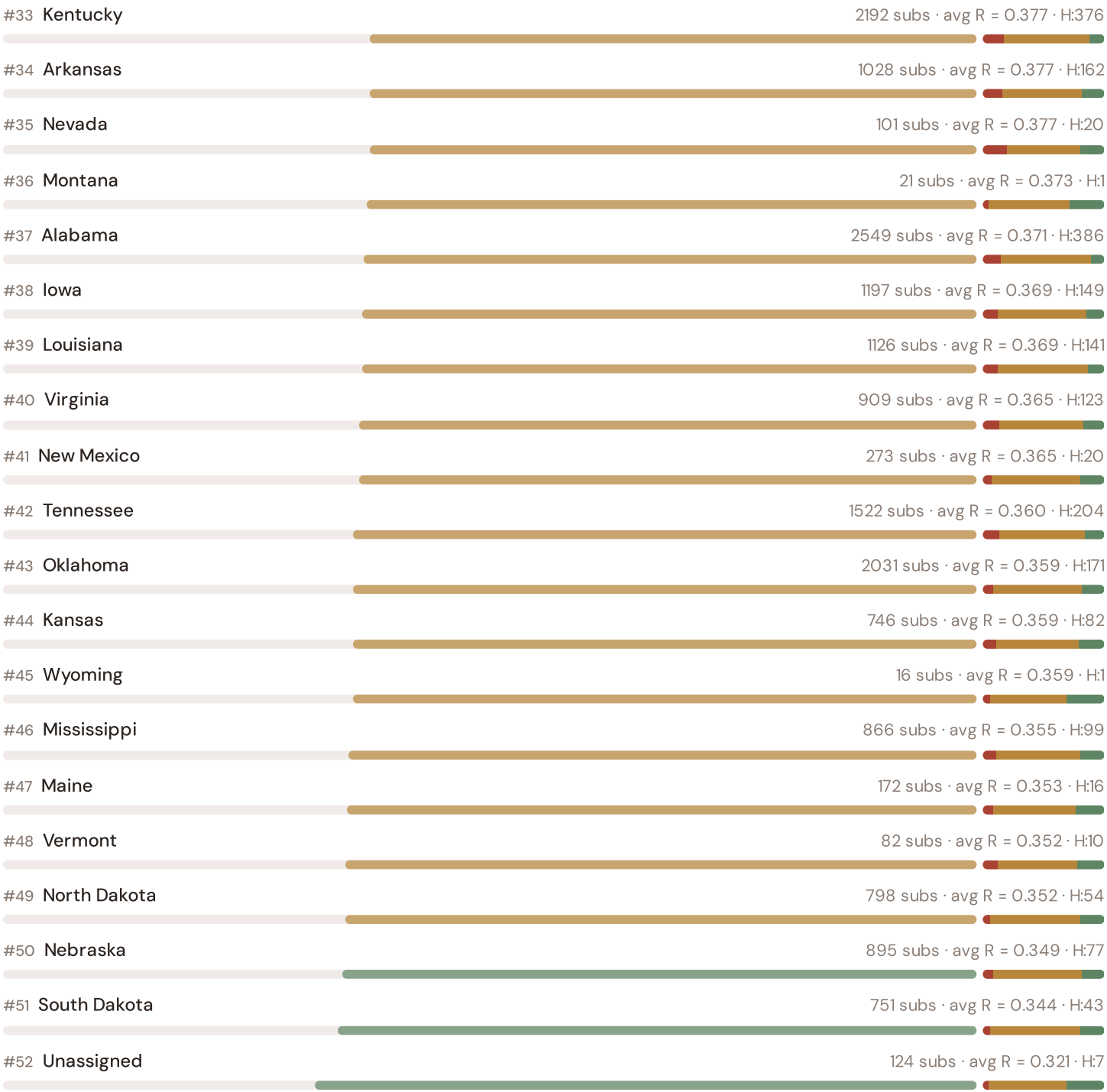
Regional Risk Ranking — All 52 States

Where does the California sit within the national landscape? Regions sorted by mean R<sub>median</sub>, with band distribution and fleet comparison.

Longer bar = lower R = better resilience. Sorted worst → best.







The California ranks **#17 of 52 states** by mean R<sub>median</sub>, placing it among the upper tier of national risk. The three most stressed regions are Washington, New Jersey, Maryland, while the three most resilient are Unassigned, South Dakota, Nebraska. 24 of 52 states score above the fleet average of 0.400. This ranking — possible only because the SSI scores every substation individually rather than relying on aggregate DNO or national statistics — reveals the true spatial distribution of grid stress across UK. It is precisely this substation-level granularity that positions the SSI as a tool for Net Zero Strategy / Levelling Up Fund investment prioritisation: not treating entire regions as monoliths, but identifying the specific corridors within each that demand attention.

SECTION F — RECURRING MODULE

## SSI Enhanced Neural Network — Monthly Topic

**From grid intelligence to BESS valuation platform.** The SSI Index answers *"How healthy is this substation?"* The SSI Enhanced Neural Network (SSI-ENN) extends this into: *"What is a BESS worth at this substation — to the NGESO/DNO and to the community?"* Each edition of this report explores one building block of the SSI-ENN's five-layer architecture, showing how the SSI Index data feeds directly into investment-grade grid analytics.

#### Layer 1 — Data

SSI v4.0 component taxonomy, normalisation, MC engine

#### Layer 2 — Valuation

NGESO/DNO + Community value stacks, TCO, real options

#### Layer 3 — Neural Net

GBT+DNN ensemble, transfer learning, SHAP

#### Layer 4 — Coverage

Country prioritisation, DCI, registry, calibration

#### Layer 5 — Platform

Products, API, revenue model, unit economics

#### LAYER 3

§0.5.5 · edition · May 2026

## Uncertainty Quantification

*Investment-grade predictions require calibrated confidence intervals, not just point estimates*

A point estimate without uncertainty bounds is unusable for investment-grade analysis. The SSI-ENN produces calibrated P10/P50/P90 prediction intervals using two complementary approaches: quantile regression (the DNN head directly outputs multiple quantiles, trained with pinball loss) and conformal prediction (a distribution-free method that provides guaranteed coverage). The SSI's own Monte Carlo confidence intervals (CI\_width, P5/P95) serve as the ground truth for calibration: the neural network's uncertainty estimates are validated against the analytical uncertainty from the 10,000-iteration copula engine. If the NN says "P90 NPV = £4.2M" for a substation, that bound must be calibrated — meaning 90% of true valuations (from the full analytical model) should indeed fall below £4.2M. This calibration is verified per country, per voltage class, and per risk band.

#### KEY CONCEPTS

- Quantile regression (pinball loss) + conformal prediction
- Calibrated against SSI Monte Carlo ground truth
- Per-country, per-voltage, per-band calibration verification
- Investment-grade: P10/P50/P90 for all outputs

#### LIVE SSI DATA CONNECTION

The American training set comprises 45,003 substations with complete 95-feature vectors. Average CI\_width of 0.147 across the fleet provides the uncertainty ground truth that the neural network's quantile regression heads must calibrate against. The fleet's risk distribution — 4720 Low, 29773 Medium, 10507 High, 3 Critical — ensures balanced training across all risk bands.

**Coming next month:** **LAYER 3** Explainability & SHAP Values — *Clients need to understand not just the prediction but WHY a substation ranks highly*



# Looking Ahead

NEXT MONTH — EDITION 003

## The Yorkshire Wind Corridor

Deep-dive into Yorkshire's grid risk profile — substation map, component analysis, county breakdown. American Context Card: Ebro valley wind farms — grid absorption capacity and curtailment analysis.

NEXT DATA REFRESH

## Q3 2026 — 1 Quarterly Sources

DOE are due for refresh in Q3. Impact: BEIS/DESNZ (Renewable Energy Planning Database) data refresh schedule (T1), business fabric indicators (E2), macro-economic data. Highest-impact annual source pending: **EIA (Energy Information Administration)** (8 variables → C1–C4 (SAIDI/SAIFI/CAIDI), quality regulation).

FLEET SPOTLIGHT

## o Substations Approaching Critical Degradation

The SSI's 4-state Markov degradation model identifies **O substations** with a stationary critical probability above 70% — meaning their long-term equilibrium state is dominated by degraded or critical condition. Without intervention, these substations will trend toward failure. Average estimated time-to-critical: ? years.

Edition 003 will be published on the second Thursday of June 2026. Deep-dive region: **Yorkshire**. To ensure you receive it, confirm your subscription segment (General / Institutional / Academic) by email to [ssi\\_index@ikenga.eu](mailto:ssi_index@ikenga.eu).

**Feedback welcome.** The SSI Index is designed to evolve through stakeholder dialogue. If you represent a American utility, FERC, NGESE, regional DNO, municipal energy company, or academic institution and would like to contribute data or methodology feedback, please contact us at [ssi\\_index@ikenga.eu](mailto:ssi_index@ikenga.eu). The SSI's credibility depends on open scrutiny.

### SSI Monthly Intelligence Report — Edition 002

Published 14 May 2026 · SSI Index v4.0.2 · 45,003 substations · 22,765 grid lines · 6 components · 20 metrics · 6 modifiers  
10,000-iteration Gaussian Copula Monte Carlo · Sobol global sensitivity analysis (196,608 evals/substation) · CMIP6 SSP2-4.5